

Andrew P. Sabelhaus

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EDUCATION

PH.D. MECHANICAL ENGINEERING
M.S. MECHANICAL ENGINEERING
B.S. MECHANICAL ENGINEERING

University of California, Berkeley. 2019
 University of California, Berkeley. 2014
 University of Maryland, College Park. 2012

APPOINTMENTS

Boston University
 Carnegie Mellon University

Assistant Professor, Mechanical Engineering, 2022 - Present
 Postdoctoral Research Fellow, Mechanical Engineering, 2019-2021

AWARDS AND HONORS

4. NSF Faculty Early Career Development Program (CAREER) Award. 2024
3. Intelligence Community Postdoctoral Research Fellowship. 2020
2. NASA Space Technology Research Fellowship. 2015
1. NSF Graduate Research Fellowship. 2012

RESEARCH OUTPUT SNAPSHOT

Peer-Reviewed Publication Count:				Total Citations:	h-index:
	Conference:	Journal:	Total:		
1st-Author or PI:	11	7	18	1689	19
All:	17	14	31		

RELEVANT PUBLICATIONS

6. A. Dickson, J.C. Pacheco Garcia, A.P. Sabelhaus, "Force-Safe Environment Maps and Real-Time Detection for Soft Robot Manipulators," *IEEE International Conference on Soft Robotics (RoboSoft)*, April 2026. arXiv:2511.05307
5. L. Zamora Yanez, J. Rogatinsky, D. Recco, S.-Y. Lee, G. Matthews, A.P. Sabelhaus, D. Hoganson, T. Ranzani, "Soft Robotic Delivery of Coiled Anchors for Cardiac Interventions," *IEEE Robotics and Automation Letters*, Vol. 10, Issue 12, Dec 2025. doi:10.1109/LRA.2025.3617249
4. A. Dickson, J.C. Pacheco Garcia, M.L. Anderson, R. Jing, S. Alizadeh-Shabdiz, A.X. Wang, C. DeLorey, Z. Patterson, A.P. Sabelhaus, "Safe Autonomous Environmental Contact for Soft Robots using Control Barrier Functions," *IEEE Robotics and Automation Letters*, Vol. 10, Issue 11, Nov. 2025. doi:10.1109/LRA.2025.3609669
3. R. Jing, C. Van Hook, I. Yang, A.P. Sabelhaus, "Fault Detection and Response for Safe Control of Artificial Muscles in Soft Robots," *IEEE Control Systems Letters (L-CSS)*, Vol. 9, Sept. 2025. doi:10.1109/LCSYS.2025.3610637
2. A. Dickson, J.C. Pacheco Garcia, R. Jing, M.L. Anderson, A.P. Sabelhaus, "Real-Time Trajectory Generation for Soft Robot Manipulators Using Differential Flatness," *IEEE International Conference on Soft Robotics (RoboSoft)*, April 2025. doi:10.1109/RoboSoft63089.2025.11020810
1. A. Choi, R. Jing, A.P. Sabelhaus, M.K. Jawed, "DisMech: A Discrete Differential Geometry-Based Physical Simulator for Soft Robots and Structures," *IEEE Robotics and Automation Letters*, Feb 2024. doi:10.1109/LRA.2024.3365292